EXP NO: 30

ANALYSIS OF CELLULAR FREQUENCY PLANNING AND HANDOVER PERFORMANCE UNDER USER MOBILITY CONDITIONS USING MATLAB

Objectives

i) To understand the basic concept of cellular frequency planning and frequency reuse.
ii) To determine a suitable cellular cluster size and frequency-reuse factor for a simplified hexagonal cellular layout.
iii) To assign channel/frequency groups to cells and visualize the resulting reuse pattern.
iv) To model a mobile user moving through adjacent cells and calculate the received signal strength from serving and neighboring base stations.
v) To identify handover conditions using a received-signal-strength comparison with a handover threshold and hysteresis margin.
vi) To measure and analyze handover locations, number of handovers, outage regions, and signal strength variation under user mobility.

CODE

```
clc;
clear;
close all;

% Cell Centers (3 Cells)
cellPos = [0 500 1000];

% User Movement
user = 0:10:1000;

% Nearest Cell Selection (Handover)
handover = zeros(size(user));

for i = 1:length(user)
   [~, handover(i)] = min(abs(user(i) - cellPos));
end

% Display Results
disp('User Position (m)    Connected Cell');
disp([user' handover']);

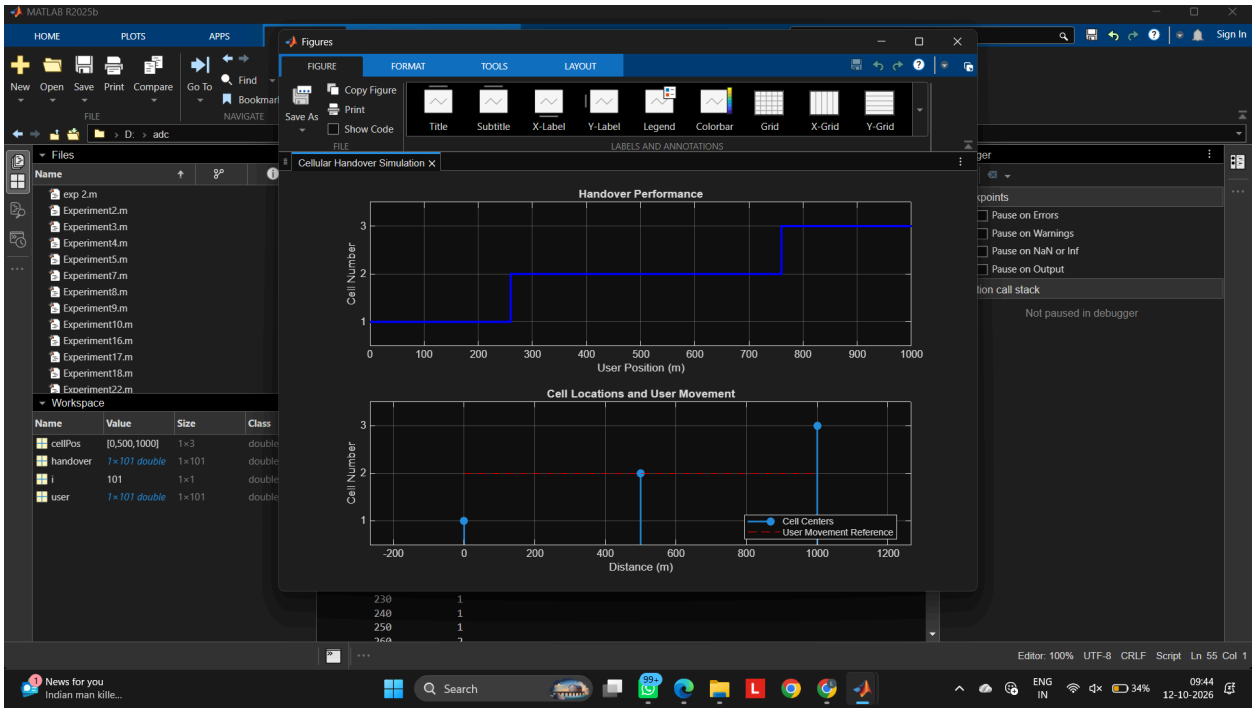
% Plot Results
figure('Name','Cellular Handover Simulation', ...
      'NumberTitle','off');
```

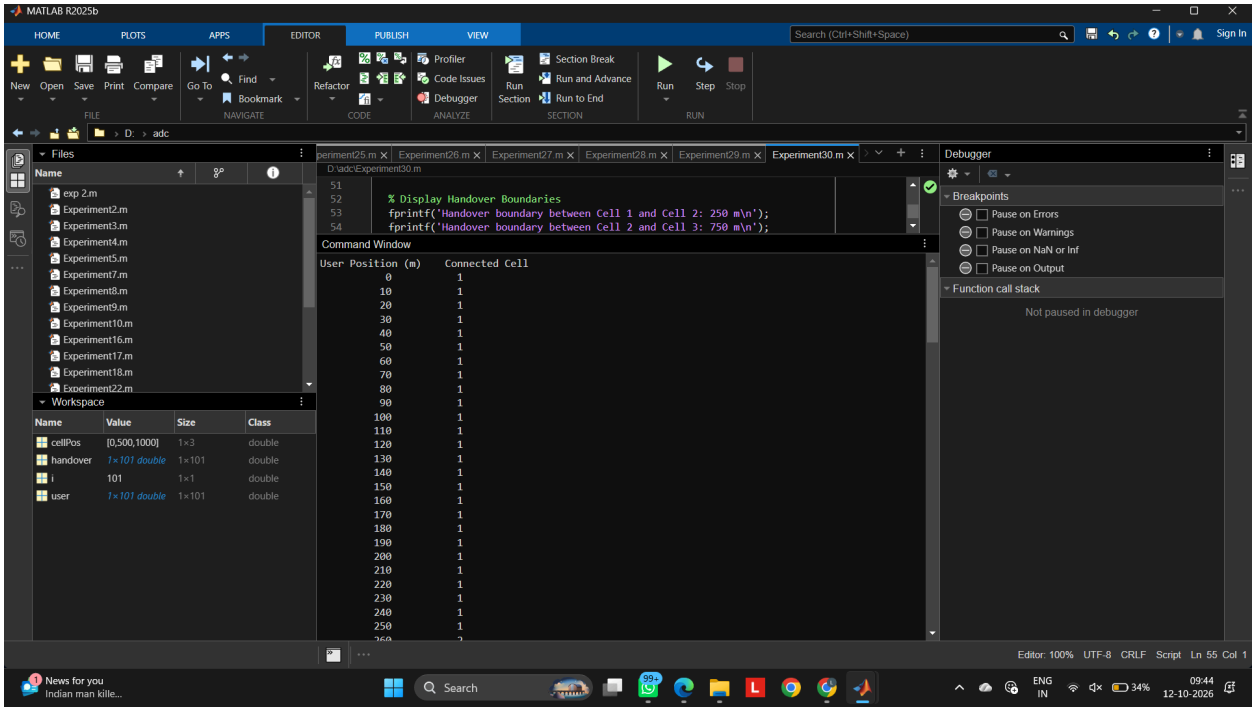
```
% Handover Performance
subplot(2,1,1);
stairs(user, handover, 'b', 'LineWidth', 2);
title('Handover Performance');
xlabel('User Position (m)');
ylabel('Cell Number');
yticks([1 2 3]);
ylim([0.5 3.5]);
grid on;

% Cell Locations and User Movement
subplot(2,1,2);
stem(cellPos, [1 2 3], 'filled', 'LineWidth', 1.5);
hold on;
plot(user, 2*ones(size(user)), 'r--');
title('Cell Locations and User Movement');
xlabel('Distance (m)');
ylabel('Cell Number');
yticks([1 2 3]);
ylim([0.5 3.5]);
legend('Cell Centers', 'User Movement Reference', ...
    'Location', 'best');
grid on;
hold off;

% Display Handover Boundaries
fprintf('Handover boundary between Cell 1 and Cell 2: 250 m\n');
fprintf('Handover boundary between Cell 2 and Cell 3: 750 m\n');
```

OUTPUT


Figures
Cellular Handover Simulation
Handover Performance
Cell Number
User Position (m)
Cell Locations and User Movement
Cell Centers
User Movement Reference
Distance (m)


% Display Handover Boundaries
fprintf('Handover boundary between Cell 1 and Cell 2: 250 m\n');
fprintf('Handover boundary between Cell 2 and Cell 3: 750 m\n');
Command Window
User Position (m) Connected Cell